VUP Bridge

A market, a payment bridge and a planner tool for households that receive VUP support

Full project description, accounts and login guide, and analysis

NISR 2026 Big Data Hackathon (draft for track choice: Track 3 Open Innovation or Track 2 Financial Inclusion and Poverty Reduction)

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Team member 2	_____ (team needs 2 students, at least one Rwandan citizen)
Date	22 October 2026
Status	Draft for review. Numbers are early. Written in easy words, with the competition facts and the NISR submission checklist.

1. Summary

Many poor families in Rwanda depend on VUP support to buy food. NISR survey data shows that payments often arrive late: the average wait is 8 to 11 days, and about half of beneficiaries wait more than a week. During that wait families can go hungry or borrow. At the same time, farmers often struggle to find buyers.

VUP Bridge is a web app that does three jobs. It is a **market** where farmers sell food directly. It is a **bridge**: when a VUP payment is late, a family can still buy food on credit, a donation pool pays the farmer first, and the late VUP payment repays the pool. It is a **planner tool** that shows government staff where and for whom payments are late, using real NISR data.

One sentence: VUP Bridge helps families eat while they wait for late VUP payments, helps farmers sell, and helps planners see where delays happen.

The app has four kinds of users: VUP member (family), seller (farmer), shopper, and planner. The planner side contains the full analysis of the NISR data. The first prototype uses fake money. Real money would need approval from the financial authorities.

How to use this document: read it from the top. Sections 1 to 5 explain the project. Section 6 has the analysis and numbers. Section 12 is the work plan. Section 14 is the NISR submission checklist. Tick each row as you finish it. Hard words are explained in the glossary at the end.

2. The problem and the evidence

The evidence comes from the NISR VUP household survey data (3,771 households, 15,039 people, 20 files). The early findings below use 3,399 beneficiary records across four VUP components.

Programme	Average delay	Paid over 7 days late	Records
Expanded Public Works	11.0 days	63%	796
Classic Public Works	9.7 days	50%	925
Direct Support	8.3 days	51%	1,003
NSDS (nutrition)	7.7 days	42%	675

Delay is the average days late of the last payments, as reported by the household. Early numbers. They are simple counts and are not yet adjusted to match the whole country.

- Overall: average delay 9.2 days, about 52% paid more than 7 days late.
- The Southern and Eastern provinces wait longest (10.7 days). The Northern Province waits least (6.6 days).
- The biggest hotspot is Expanded Public Works in the Southern Province: about 14.6 days, and 79% paid over 7 days late.
- Delays are short and common (only 1 to 2% are over 30 days), so the bridge should cover about one to two weeks.

Important rule: the data shows patterns. It cannot show why a payment was late. The app says this on every analysis page.

3. The idea: how the app works

Example story: Amina receives Direct Support. Her payment is 8 days late.

Step	What happens
1	A farmer lists beans at 780 RWF per kg. The app compares this with the NISR median (800 RWF) and shows 'Fair'.
2	Amina's payment is late. The app opens a small credit limit: people in the family x daily food cost x days late.
3	Amina buys the beans on credit.
4	The farmer delivers. The donation pool pays the farmer straight away.
5	Shoppers keep the pool filled: a small share (for example 2%) of what they spend goes into it.
6	The VUP payment arrives. Amina's credit is repaid to the pool.
7	The planner sees credit given, credit still open, the pool balance and the NISR delay patterns.

4. Users and accounts

4.1 The four account types

Account type	Who they are	What they can do
VUP member (family)	A household that receives VUP support	See payment status, get budget meal ideas, buy food, use bridge credit when the payment is late
Seller (farmer)	A farmer, cooperative or small food seller	Add listings, see price guide, get orders, mark delivered, see payment status
Shopper	Anyone who can afford to shop	Buy food as normal. A share of each purchase goes to the pool. See how many days of food they funded
Planner	Government or programme staff (for example district or LODA staff)	See all dashboards and analysis, orders summary, credit and pool. No buying or selling

4.2 How to create an account

Account type	Sign-up steps	Information collected	Check (real version)
VUP member	1. Choose 'I receive VUP support'. 2. Enter name and phone. 3. Choose district and sector. 4. Choose VUP programme and household size. 5. Create a password.	Name, phone, district, sector, programme, household size	Matched against the official VUP beneficiary list, or confirmed by the Umurenge officer, before credit opens
Seller	1. Choose 'I sell food'. 2. Enter name, phone and district. 3. Choose what you grow or sell. 4. Add a mobile money number for payment. 5. Create a password.	Name, phone, district, produce, mobile money number	Confirmation by a cooperative or local leader. Mobile money number checked
Shopper	1. Choose 'I want to buy'. 2. Enter name and phone or email. 3. Choose the share to give to the pool. 4. Create a password.	Name, phone or email, donation share	Phone or email code
Planner	No self sign-up. An administrator sends an invitation to an official email. The planner sets a password from the link.	Name, institution, official email, role	Official email and administrator approval

In the hackathon prototype all checks are skipped and accounts are demo accounts.

4.3 Sample logins (demo only)

These are made-up people and made-up passwords, to test the prototype. Never reuse them in a real system.

Role	Full name	Username	Password	Details
VUP member	Amina Uwase	amina_family	Amina#2026	Direct Support, 5 people, Kirehe
VUP member	Jean Bosco Habimana	bosco_family	Bosco#2026	Expanded Public Works, 6 people, Gisagara
Seller	Mukamana Farm	mukamana_seller	Farm#2026	Dry beans, Kirehe
Seller	Nyabihu Potato Cooperative	nyabihu_seller	Coop#2026	Irish potato, Nyabihu
Shopper	Grace Ingabire	grace_shopper	Grace#2026	Gives 2% to the pool
Shopper	Eric Mugisha	eric_shopper	Eric#2026	Gives 3% to the pool
Planner	Planner Demo	planner_demo	Plan#2026	Sees all dashboards

4.4 Security and privacy rules for the real version

- Passwords are never saved as readable text. They are saved in a scrambled form that cannot be read back. The sample passwords above exist only in the demo.
- Each user sees only their own data. Families do not see other families. Planners see summaries, not names.
- Collect only what is needed. Follow Rwanda's personal data protection law (Law No. 058/2021) and get legal advice before launch.
- Household rows from the NISR survey are never published on the website or on GitHub. Only summary tables are.
- Poor and non-poor users see the same screens, so no one is marked out.

5. What each user sees

Role	Screens
VUP member	Payment status (programme, days late). Bridge credit left. Basket cost per day for the household. Market with price guide. Buy now or buy on credit. My orders. Budget meals.
Seller	Add listing (food, district, price per kg, quantity). Price guide: Low below 85% of the NISR median, High above 115%. Orders to deliver. Mark delivered. Payment status.
Shopper	Market. Buy food. Share to the pool (1 to 5%). Gifts so far and days of food covered. My orders.
Planner	Overview stats (orders, credit given, credit open, pool balance). Pool flow. A button to simulate VUP payments arriving. The full analysis section (Part 6). Recommendations.

6. The analysis part

The analysis is built into the planner side, and its main findings are also shown to other users in simple form. All numbers are early. They are simple counts. The final run will adjust them to match the whole country, test if the differences are real, and show how sure we are.

A1. Payment delays by place

Province	Average delay	Paid over 7 days late	Records
Southern	10.7 days	64%	675
Eastern	10.7 days	59%	762
Western	9.4 days	55%	789
City of Kigali	8.3 days	47%	358
Northern	6.6 days	34%	815

Slowest districts	Avg days (records)	Fastest districts	Avg days (records)
Kirehe	15.5 (70)	Nyabihu	4.5 (63)
Gisagara	14.9 (77)	Musanze	5.1 (70)
Karongi	13.2 (100)	Gicumbi	6.3 (297)
Kicukiro	12.4 (27)	Rwamagana	6.3 (55)
Gatsibo	12.2 (92)	Rubavu	6.6 (76)

District results with few records can mislead. The app flags groups under 100 records.

A2. Delays by group of beneficiaries

Group	Average delay	Paid over 7 days late	Records
Severely poor	11.0	62%	174
Moderately poor	9.7	56%	875
Not poor	8.9	50%	2,350
Mobile money payments	10.4	59%	630
Umurenge SACCO payments	7.7	43%	546
Rural / Urban	9.3 / 8.5	52% / 50%	2,895 / 504
Women / Men	9.2 / 9.2	52% / 52%	2,468 / 931
Age 40-59 / 60 and over	9.8 / 8.9	54% / 53%	1,115 / 1,317

Poverty is measured after support is received. The link between poverty and delay is not proof of cause. Women and men show no clear difference.

A3. Hotspots: programme by province (average days late)

Programme	Southern	Eastern	Western	Kigali	Northern
Expanded Public Works	14.6	10.8	11.0	9.6	8.5
Classic Public Works	11.4	12.0	10.0	8.5	6.3
Direct Support	9.7	9.2	9.2	7.4	6.0
NSDS	8.3	11.4 (57 records)	7.8	no data	6.2

A4. Poverty risk model

Goal: estimate which households are likely to be poor, so planners know where to look first. It uses housing and household facts only. It leaves out every spending, consumption and income variable, because poverty is calculated from those. A first try wrongly counted a technical survey number (the survey weight) as a strong sign of poverty. It is not a real sign, so we removed it.

Item	Early result
Poor households in the sample	29% (3,771 households)
Methods	Two standard computer methods. Each was tested on households it had not seen before
Accuracy score (AUC)	About 0.76 to 0.79. A score of 1.0 is perfect and 0.5 is a coin flip, so this is fair but not great
Household size	Poor households 4.9 people, not poor 3.6
Main factors	Household size, estimated rent, number of children, outside floor area, floor material

Fair accuracy. Use it to decide where to look first. Never use it to decide who gets help. Poverty rates in this sample: Southern 41%, Western 36%, Northern 22% (extreme poverty 8.8%, 7.9%, 2.3%). Other provinces come in the final run.

A5. Finance access

Question	Early result (survey records)
Accounts reported	Mobile money 4,699. SACCO 4,641. Commercial bank 444
Tried to get a loan	353 of 5,404 people
Where loans come from	Tontine 2,129. Relatives 1,246. VUP financial loan 614
Why loans were refused	Purpose not clear 113. Income too low 89. No collateral 35

Counts are survey records, not national totals. The final run will adjust them to match the whole country.

A6. Food prices and where households buy

Food (per kg)	Middle price (RWF)	Food (per kg)	Middle price (RWF)
Sweet potato	250	Local rice	1,200
Cooking banana	300	Onion	1,667
Salt	400	Groundnut flour	2,000
Avocado	400	Small dry fish	3,030
Irish potato	450	Dry beans	800
Tomato	500	Corn flour (factory)	1,000

Where households buy: small shop 69% (10,957 records), market 27% (4,233), individual seller 5% (734). Individual sellers are the closest match to direct farmer sales.

Demo food basket (assumed amounts, not NISR): per person per day 120 g beans, 300 g potato, 80 g tomato, 30 g onion. With sweet potato it costs about 261 RWF per person per day. NISR has no nutrient values or recipes, so the app shows cost only and makes no calorie claims.

A7. Planned analyses (not yet done)

- **District priority score:** combine poverty rate, average payment delay and local food price level, then rank districts into a 'help here first' list or map.
- **Poverty rates by province and district** with small-sample warnings.
- **Farm-gate versus shop price gap** from the own-produced food price field.
- **Final checks:** adjust numbers to match the whole country, test if each difference is real, and show how sure we are, (simple statistical tests).
- **Poverty what-if form** returning a risk band, not a yes or no.
- **Recommendations page:** four or five actions, each linked to a goal in NST2 or Vision 2050.

7. Data and method

NISR file	Used for
Direct Support, Classic Public Works, Expanded Public Works, NSDS files	Payment channel and days late
Household file	Province, district, urban or rural, poverty group, housing fields, weights
Person file	Sex, age, household size, children
Savings and credit files	Finance access
Food expenditure file (large, selected columns only)	Prices per kg, place of purchase

- Delay per record = average of the delay fields for that person and programme. Over 7 days late = delay above 7.
- Groups under 100 records are flagged. Groups under 25 are hidden or warned.
- The public site and GitHub hold only summary tables. No household rows.

8. What is real, simulated and assumed

Type	What it covers
Real NISR analysis	Delays, hotspots, poverty model, finance access, food prices, where households buy
Simulated (demo)	Farmers, listings, orders, credit, pool, shoppers and all money (fake money)
Assumed by the team	Daily food amounts per person, shopper spending (20,000 RWF a month), donation share, credit window

9. Web app plan: all devices and Google

- **A web app that works on phones, tablets and computers** with no install. The layout fits any screen, light files for mobile data, big buttons.
- **Can be added to the phone home screen** like an app (this is called a PWA). iPhone allows fewer offline features than Android.
- **Findable on Google:** host on a public address (for example Vercel), give the public home page a clear title and description, and ask Google to look at the site (a free tool called Google Search Console). Listing can take days or weeks, so it is a bonus. Pages behind a login are private and will not appear.
- **Future:** USSD or SMS for basic phones (users dial a code like *123# and need no internet), with network operators and mobile money. Not built now.

10. Limits and risks

- The data shows patterns, not reasons. Delays are reported by households, not official payment records.
- The poverty model is moderate. Poverty is measured after support is received, so findings are links and not causes.
- NISR has no real farmer records, nutrient values or recipes.
- Holding or moving real money needs approval from the financial authorities. The submission must call this a prototype.
- Farmers could wait if the pool is empty. The design needs a clear rule on who guarantees payment.

- Small district samples. Province results are stronger than district results.

11. Hackathon fit

Criterion (20 points each)	How VUP Bridge answers it
Problem understanding and relevance	Late VUP payments and food access link to poverty and food security goals in NST2 and Vision 2050
Data use and methodology	Real NISR files, joined and analysed, with a model, weights and tests in the final run
Tech innovation	Bridge and pool design, price guide, poverty risk model, priority score
Usability and design	Four clear role views, mobile friendly, simple wording
Tangible impact	Planners can find hotspots. Families and farmers get a direct way to trade while waiting

- **Deadline:** 30 October 2026, 11:59 PM. Plan to submit by 29 October. Registration deadline was 20 October, so confirm the registration is complete.
- **Submit:** a public GitHub link, a deployed app link and documentation. Declare AI use. NISR takes the intellectual property.
- **Team:** 2 students, at least one Rwandan citizen, with student cards.

12. Status and work plan

Item	Status
Analysis demo website (delays, poverty risk, finance access, meals, bridge slider)	Built (early numbers)
Marketplace prototype (four roles, listings, orders, credit, pool)	Built (no login, saved in one browser only)
Login and account creation (this document, section 4)	To build
One combined app with the analysis inside the planner role	To build
District priority score and recommendations	To build
Final weighted numbers with tests	To run
Shared database so different people see the same orders	To build
Documentation, README and AI-use declaration	To write
Public GitHub repository with the layout in section 14.1	To do
Demo video, 3 minutes at most, recorded on a computer	To do
Student proof for both team members	To do

Days	Work
Now to 24 Oct	Data run: weights, tests, priority score, poverty by district. Create the GitHub repo and push the first version
24 to 26 Oct	Build combined app with logins and roles. Add the district map
26 to 27 Oct	Deploy, test on phones and computer, test all sample logins
27 to 28 Oct	README, documentation, declaration and AI-use statement. Record and export the demo video
29 Oct	Final test (section 14.6), upload everything, submit

13. Competition facts in easy words

These facts come from the hackathon notice. Keep them close, because they decide what we build and when.

The three tracks

Track	In easy words	Does VUP Bridge fit?
1. Agricultural Productivity	Help farmers grow, store and sell more, using data	Partly: our farmer market helps farmers sell
2. Financial Inclusion and Poverty Reduction	Understand who is left out of money services, how poverty changes, and whether help programmes work	Yes: we study VUP, a help programme, and late payments
3. Open Innovation	Pick any national problem and build a useful tool, using NISR data	Yes: our app mixes food, farming, money and social help

You choose one track. We lean to Track 3. Track 2 also fits.

Important dates (2026)

Date	What happens
1 September	Competition opens
20 October, 11:59 PM	Registration closes. Confirm you registered
30 October, 11:59 PM	Submission deadline. Plan to submit by 29 October
1 to 5 November	NISR reviews and judges the projects
13 November	Shortlisted teams announced
16 November to 4 December	Shortlisted teams get coaching, workshops and a final presentation
11 December	Awards ceremony

Prizes

- The best laptops go to the first three winning teams.
- Final-year Rwandan students in winning teams may be considered for professional internships after their studies. This depends on free places, and the field of study must match the job.
- Final-year students and women are encouraged to apply.

Who can join, and who cannot

- Undergraduate or master's students in any field of study.
- You need a student card or another official document.
- Teams have 2 students, and at least one must be a Rwandan citizen. They can be from the same or different universities.
- Past winners (1st, 2nd or 3rd place in any earlier edition) cannot join.
- The work must be your own, never sent to another competition, with all sources credited.
- You may use AI tools as helpers, but you must say so, and you are fully responsible for the work.

Getting the NISR data

Step	What to do
1	Open the NISR Microdata Portal and click Register. Fill in your details
2	The portal says a confirmation email was sent. Ignore this and log in with the details you chose
3	Search for the survey you want and click on it
4	Click Get Microdata and send the short access form
5	For public datasets, access is usually given at once. Download the zip file

We already have the VUP survey files, so this is only needed if you want more surveys.

Contact

Questions about the competition: prosper.ayinebyona@statistics.gov.rw or 0786 872 057. NISR general line: +250 788 383103.

14. Submission checklist (from the NISR preparatory webinar)

NISR gave a preparatory webinar. This section turns its slides into a checklist for our project. Anything not done by the deadline can cost points or lead to disqualification, so treat every row as a task.

Requirement	What NISR says	Our action	Status
Public GitHub link	Submit a public GitHub repository	Create a public repo (for example vup-bridge), add all files, check it opens in a private browser window	To do
Deployed app link	Must be publicly accessible	Deploy on a free host. Open the link on a phone and a laptop without logging in to any host account	To do
Sample logins	If login is required, give sample credentials so users log in easily	Use the demo accounts in section 4.3. Show them on the login page and in the README	Planned
Documentation	Submit documentation with the links	This PDF is the base. Add final numbers, the README and the declaration	Draft
README file	Explain what data the app uses, what it communicates, and how users interact with it	Write it using the outline below	To write
Script(s)	Code for the solution that works on any machine	Clear folders, a requirements file, run steps. Test on a clean computer	To do
Demo video	A .webm or .gif of the interface and functions, 3 minutes at most. Record with screen recording software. Do not record with a phone unless the solution is a mobile app	Record on a laptop with a browser screen recorder such as Screencastify. Use the script below. Export as .webm or .gif	To do
Data file and link	Include the data you use and a link to it. If files are too large, give only the link and say so in the README	Put summary tables (CSV) in the data folder. Link to the NISR Microdata Portal. State in the README that the raw survey files are not uploaded	To do
Student proof	Student card or other official document	Prepare a clear scan or photo for each team member	To do

Requirement	What NISR says	Our action	Status
Team	2 students, at least one Rwandan citizen	Name team member 2 and confirm the citizenship rule	Open
Originality and sources	Declare the work is your own and all sources are acknowledged. Failure means disqualification	Add the declaration below. List every data source, tool and library	To write
AI use	AI tools may be used as assistants, but their use must be disclosed. Failure to do so means disqualification (competition notice)	Add an AI-use statement to the README and the documentation	To write
Intellectual property	All rights in the submitted work pass to NISR	Read and accept this before submitting. Do not include material you cannot give away	Noted
Deadline	30 October 2026, 11:59 PM	Submit by 29 October	Planned

Judging criteria on the slides are the same five areas listed in section 11.

14.1 Repository layout (at least 4 files)

NISR asks for at least four things: a README, the script(s), the demo video, and the data file with a link. The sample layout in the slides is README.md, app.py or app.R, a data folder, a demo folder with demo.webm, and requirements.txt. Ours:

Path	What it holds
README.md	What the data is, what the app shows, how to use it, how to run it, logins, AI-use statement
app/	The web app files (market, bridge, planner and analysis pages)
scripts/	Analysis scripts: clean and join files, delay tables, poverty model, finance and food tables, priority score
data/	Summary tables (CSV) and data_sources.md with the NISR Microdata Portal link. No household rows
demo/demo.webm	Screen recording, 3 minutes at most
requirements.txt	Libraries needed to run the scripts
DECLARATION.md	Originality and sources statement, and AI-use statement

14.2 README outline

- **Title and one-line summary** of VUP Bridge.
- **Problem** in three or four sentences, with the key NISR numbers.
- **Data:** which NISR files, the portal link, and a note that raw files are not uploaded.
- **What the app shows:** each role and each analysis page, and what message it should communicate.
- **How to use it:** deployed link, sample logins, and what to click for each role.
- **How to run it on your own machine:** install steps, run steps, and how to rebuild the summary tables.
- **Method and limits:** how delays and the poverty model are calculated, and the 'patterns not reasons' rule.
- **Real, simulated and assumed parts**, as in section 8.
- **AI-use statement, sources and team.**

14.3 Demo video script (3 minutes at most)

Time	Show
0:00 to 0:20	The problem: late VUP payments, with one key NISR number
0:20 to 1:00	Planner login: delays by programme and province, hotspots, poverty risk
1:00 to 2:15	The full flow: farmer adds a listing, a family with a late payment buys on credit, the pool pays the farmer, the VUP payment repays the pool
2:15 to 2:45	Shopper gives to the pool. Show the account types and sample logins
2:45 to 3:00	Limits (prototype, fake money, patterns not reasons) and the impact for planners, families and farmers

Record on a computer, not a phone. Practise once first. Keep the file under 3 minutes.

14.4 Declaration template (edit before use)

Originality and sources. We declare that this submission is our own original work and has not been submitted to any other competition or entity. All sources are acknowledged: NISR VUP household survey data (NISR Microdata Portal), and the open-source libraries listed in requirements.txt.

AI use. We used an AI assistant (Claude, by Anthropic) to help plan the project, analyse data, and write code and documentation. We reviewed and tested the results, and we take full responsibility for the submitted work.

Names, registration numbers and date: _____

Check that the AI-use statement matches what you really did before you sign it.

14.5 Webinar hints we will follow

- **Use other relevant datasets, for example shapefiles for maps.** Add a district map of the priority score and delays. Use a public Rwanda district boundary file (the NISR portal has a Geodata section) and cite the source.
- **Functionality over fancy looks.** Judges want the story the data tells. Keep the design plain and clear.
- **Do not wait for the last minute.** Deploy a first working version early, then improve it.
- **Dashboard tools such as Shiny, Dash or Streamlit are suggestions.** Any language is allowed, so our web app is fine. The README must still explain how to run it.
- **Use GitHub from the start** (create repo, add files, commit, push), not only on the last day.

14.6 Final test before submitting

- Open the deployed link in a private window on a phone and a laptop. No host login should be needed.
- Log in with every sample account. Each one works and shows the right screens.
- The repo is public and has the README, scripts, data folder, demo video and requirements file.
- The video plays, is under 3 minutes, and was recorded on a computer.
- The README steps work on a clean computer.
- Declaration, AI-use statement and student proof are ready.

15. Decisions still needed

- Track: Track 3 (Open Innovation) fits best. Track 2 also fits.
- Pool share: 2%, or another number?

- Who gets pool help first: longest delay, lowest poverty level, or both?
- Language: English only, or English and Kinyarwanda?
- App name: VUP Bridge (the analysis idea was first called VUP Pulse).
- Who is team member 2?

Glossary

Term	Meaning
VUP	Vision 2020 Umurenge Programme. Rwanda's social protection programme
NISR	National Institute of Statistics of Rwanda
NST2	Second National Strategy for Transformation
Vision 2050	Rwanda's long-term development plan
NSDS	Nutrition-Sensitive Direct Support, for pregnant women and mothers of young children
SACCO	Savings and credit cooperative
Tontine	A rotating savings group
PWA	Progressive web app: a website that can be installed on a phone
AUC	A score from 0.5 to 1.0 for how well a model tells poor from not poor
Model	A computer method that learns from data to make an estimate
Hotspot	A place or group where delays are worst
Middle price (median)	The price in the middle when all prices are put in order. It is not changed much by a few very high or very low prices
Priority score	One number that ranks places by how much help they need
Prototype	A first working version made to test the idea. It uses fake money
Simulated	Made up for the demo, but built to look like real life
Adjusting to the whole country (weights)	Survey answers are scaled up so they match all of Rwanda, not only the households surveyed
Test (statistical)	A check on whether a difference is real or could just be luck
Repository (repo)	The online folder on GitHub that holds your project files
Deployed app	The app placed online so anyone can open it with a link
USSD	Dialling a code like *123# on any phone, with no internet
Microdata Portal	NISR website where students download survey data